

Unit-3: Evaluating Models

Lesson Title: Evaluating Models	Approach: Session + Activity
Summary: In this module, students are introduced to the common metrics used to evaluate AI models. They will know how to derive and calculate the evaluation metrics and would also get an idea on how to improve the accuracy/efficiency of an AI Model. They will be introduced to the concept of Train/ Test Split, Common evaluation metrics such as Accuracy, Confusion Matrix, Precision, Recall, F1 Score) Learners will also be able to identify the use of this metrics in use cases encountered in everyday life.	
Learning Objectives: 1. To introduce students to the common metrics used to evaluate AI models 2. To familiarize students with deriving and calculating the evaluation metrics 3. To enable students to recognize the most suitable evaluation metric for a given application.	
Learning Outcomes: 1. Recognise common metrics used to evaluate AI models 2. Derive and calculate the evaluation metrics 3. Recognize the most suitable evaluation metric for a given application.	
Pre-requisites: Essential understanding of Artificial Intelligence	
Key-concepts: 1. Importance of model evaluation 2. Evaluation metrics for classification	

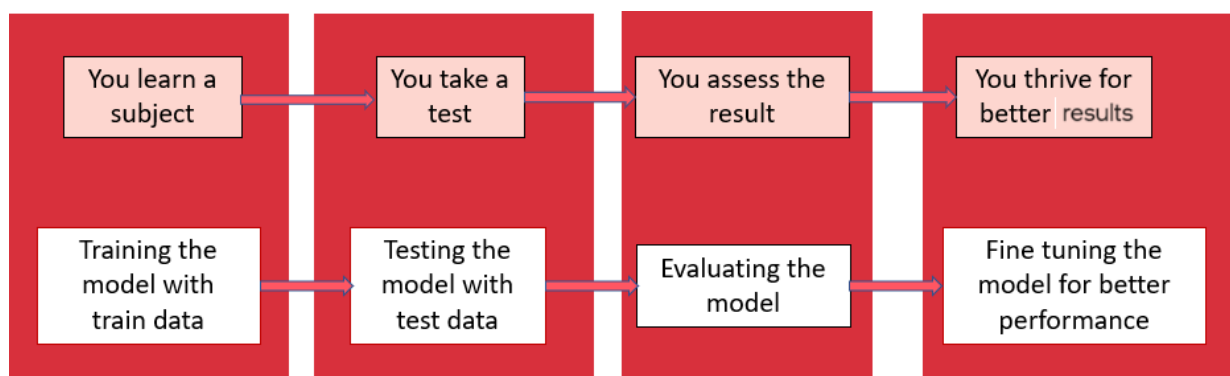
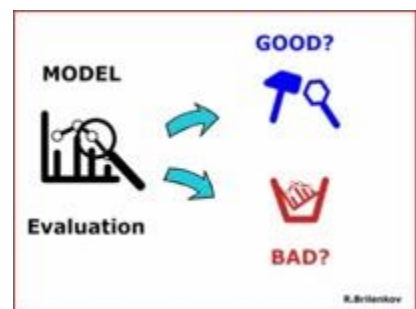
Introduction

Till now we have learnt about the 4 stages of AI project cycle, viz. Problem scoping, Data acquisition, Data exploration and modelling. While in modelling we can make different types of models, how do we check if one's better than the other? That's where Evaluation comes into play. In the Evaluation stage, we will explore different methods of evaluating an AI model. Model Evaluation is an integral part of the model development process. It helps to find the best model that represents our data and how well the chosen model will work in the future

3.1: Importance of Model Evaluation

What is evaluation?

- Model evaluation is the process of using different evaluation metrics to understand a machine learning model's performance
- An AI model gets better with constructive feedback
- You build a model, get feedback from metrics, make improvements and continue until you achieve a desirable accuracy



- It's like the report card of your school
- There are many parameters like grades, percentage, percentiles, ranks
- Your academic performance gets evaluated and you know where to work more to get better



Need of model evaluation

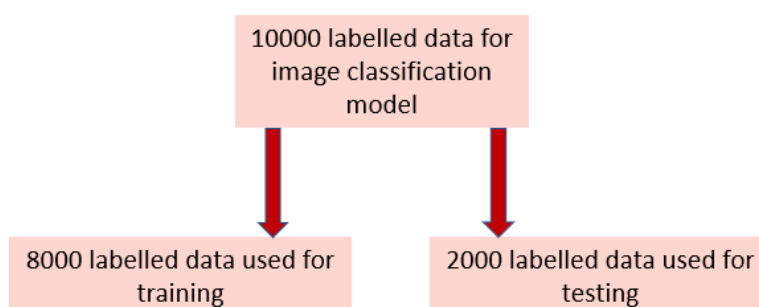
In essence, model evaluation is like giving your AI model a report card. It helps you understand its strengths, weaknesses, and suitability for the task at hand. This feedback loop is essential for building trustworthy and reliable AI systems.

After understanding the need for Model Evaluation, let's know how to begin with the process. There can be different Evaluation techniques, depending of the type and purpose of the model.

3.2: Splitting the training set data for Evaluation

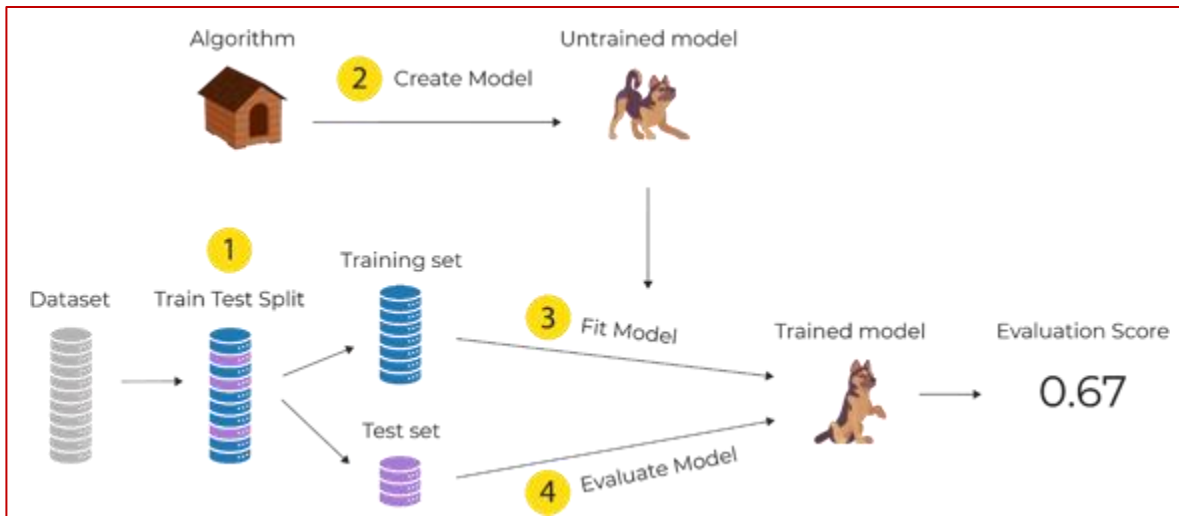
Train-test split

- The train-test split is a technique for evaluating the performance of a machine learning algorithm
- It can be used for any supervised learning algorithm
- The procedure involves taking a dataset and dividing it into two subsets: The training dataset and the testing dataset
- The train-test procedure is appropriate when there is a sufficiently large dataset available



Need of Train-test split

- The train dataset is used to make the model learn
- The input elements of the test dataset are provided to the trained model. The model makes predictions, and the predicted values are compared to the expected values
- The objective is to estimate the performance of the machine learning model on new data: data not used to train the model



This is how we expect to use the model in practice. Namely, to fit it on available data with known inputs and outputs, then make predictions on new examples in the future where we do not have the expected output or target values.

Remember that It's not recommended to use the data we used to build the model to evaluate it. This is because our model will simply remember the whole training set, and will therefore always predict the correct label for any point in the training set. This is known as **overfitting**.



You will learn more about the concepts including train test split and cross validation in higher classes.

3.3: Accuracy and Error

- Bob and Billy went to a concert
- Bob brought Rs 300 and Billy brought Rs 550 as the entry fee for that
- The entry fee per person was Rs 500
- Can you tell:
 - Who is more accurate? Bob or Billy?
 - How much is the error for both Bob and Billy in estimating the concert entry fee?



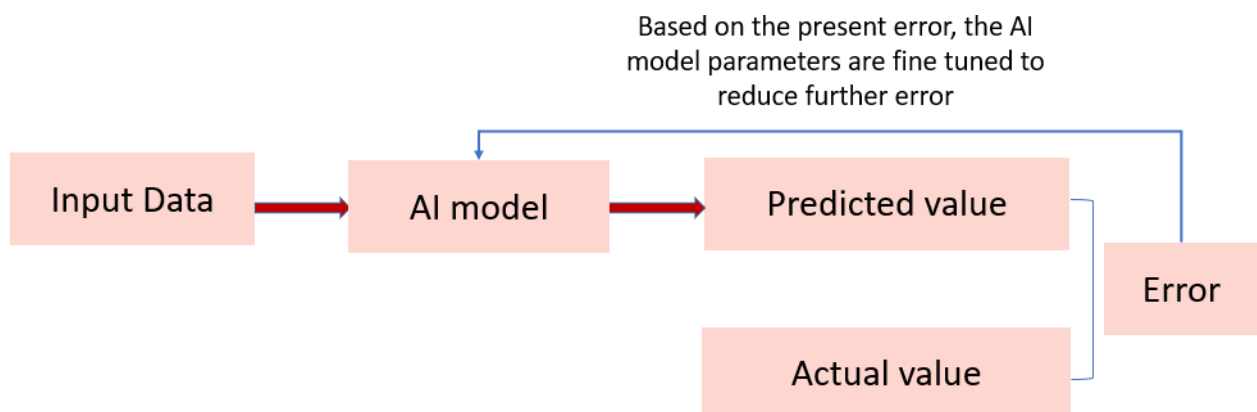
Accuracy

- Accuracy is an evaluation metric that allows you to measure the total number of predictions a model gets right.
- The accuracy of the model and performance of the model is directly proportional, and hence better the performance of the model, the more accurate are the predictions.

Error

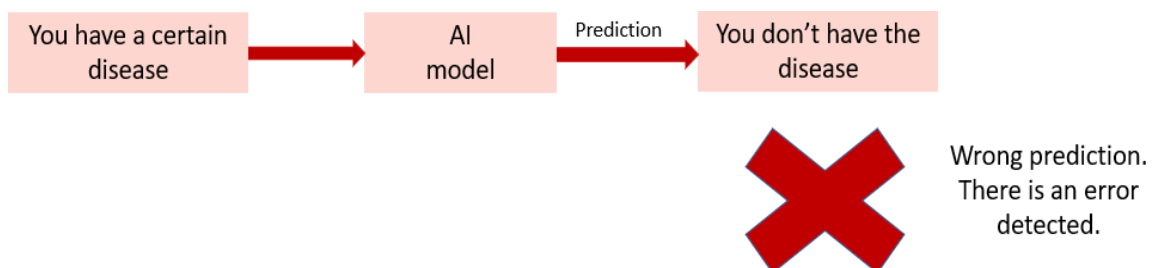
- Error can be described as an action that is inaccurate or wrong.
- In Machine Learning, the error is used to see how accurately our model can predict data it uses to learn new, unseen data.
- Based on our error, we choose the machine learning model which performs best for a particular dataset.

Error refers to the difference between a model's prediction and the actual outcome. It quantifies how often the model makes mistakes.



Imagine you're training a model to predict if you have a certain disease (classification task).

- **Error:** If the model predicts you don't have a disease but you actually have a disease, that's an error. The error quantifies how far off the prediction was from reality.



- **Accuracy:** If the model correctly predicts disease or no disease for a particular period, it has 100% accuracy for that period.

Key Points:

- Here the goal is to minimize error and maximize accuracy.
- Real-world data can be messy, and even the best models make mistakes.
- Sometimes, focusing solely on accuracy might not be ideal. For instance, in medical diagnosis, a model with slightly lower accuracy but a strong focus on avoiding incorrectly identifying a healthy person as sick might be preferable.
- Choosing the right error or accuracy metric depends on the specific task and its requirements.

Understanding both error and accuracy is crucial for effectively evaluating and improving AI models.

Activity 1: Find the accuracy of the AI model

Purpose: To understand how to calculate the error and the accuracy.

Say: “The youth will understand the concept of accuracy and error and practice it mathematically.”

Calculate the accuracy of the House Price prediction AI model

- Read the instructions and fill in the blank cells in the table.
- The formula for finding error and accuracy is shown in the table
- Accuracy of the AI model is the mean accuracy of all five samples
- Percentage accuracy can be seen by multiplying the accuracy with 100

Predicted House Price (USD)	Actual House Price (USD)	Error Abs (Actual-Predicted)	Error Rate (Error/Actual)	Accuracy (1-Error rate)	Accuracy% (Accuracy*100)%
391k	402k	Abs (402k-391k)= 11k	11k/402k=0.027	1-0.027= 0.973	0.973*100%= 97.3%
453k	488k				
125k	97k				
871k	907k				
322k	425k				

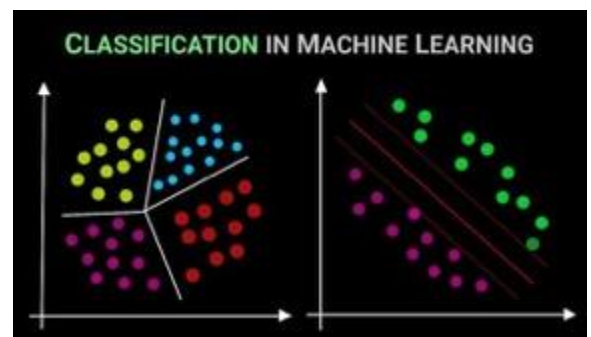
*Abs means the absolute value, which means only the magnitude of the difference without any negative sign (if any)

The Model Evaluation stands on the two pillars of accuracy and error. Let's understand some more metrics standing on these two pillars.

3.4: Evaluation metrics for Classification

What is Classification?

- You go to a supermarket and were given two trolleys
- In one, you have to place the fruits and vegetables; in the other, you must put the grocery items like bread, oil, egg, etc.
- So basically, you are classifying the items of the supermarket into two classes:
 - fruits and vegetables
 - grocery
- Classification usually refers to a problem where a specific type of class label is the result to be predicted from the given input field of data
- For example, here we are working on a vegetable-grocery-classifier model that predicts whether the item in the supermarket is a vegetable or a grocery item



Visualizing the concept of classification: Left 4 Classes; Right 2 classes

Try Yourself:

Which of this is a classification use case example?

House price prediction

Credit card fraud detection

Salary prediction

Classification Metrics

Popular metrics used for classification model

- Confusion matrix
- Classification accuracy
- Precision
- Recall

Let's understand these metrics in details:

Confusion matrix

Let's say, based on some clinical parameters; you have designed a classifier that predicts whether a person is infected with a certain disease or not.

The output is 1 if the person is infected or 0 if the person is not infected. That is, 1 and 0 signify whether a person is infected or not.

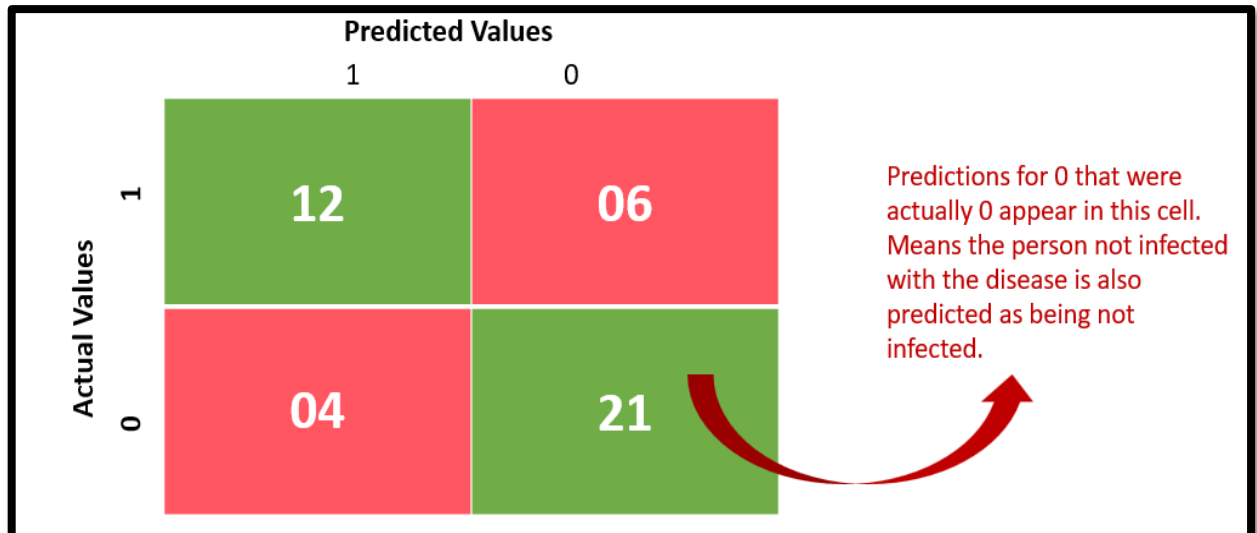
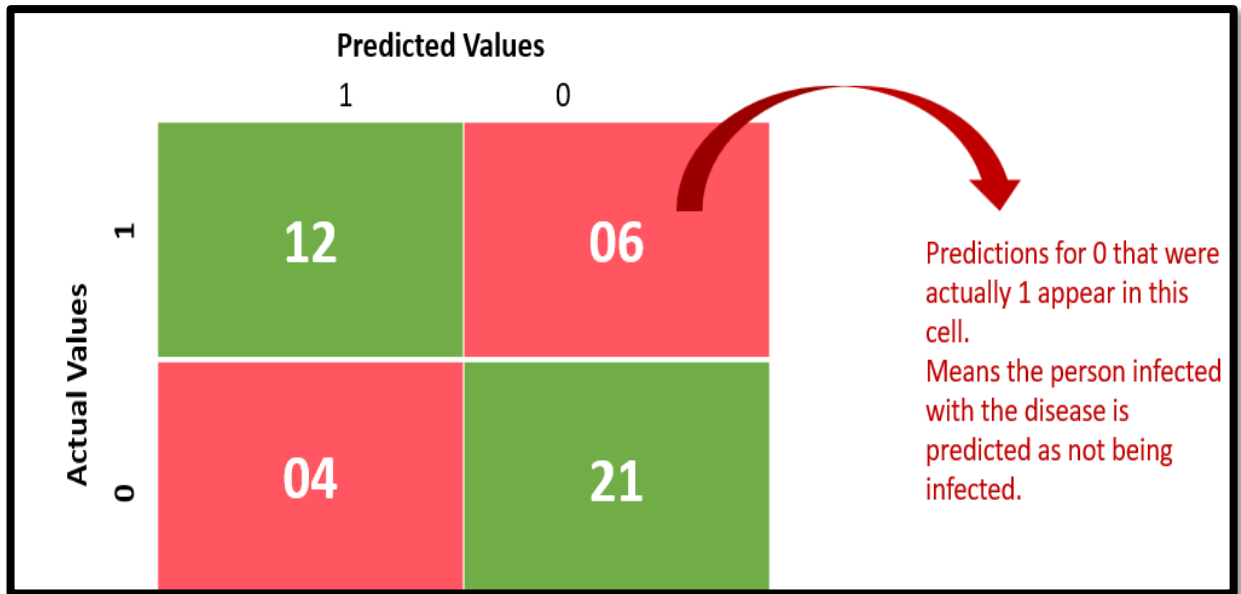
- The confusion matrix is a handy presentation of the accuracy of a model with two or more classes
- The table presents the actual values on the y-axis and predicted values on the x-axis
- The numbers in each cell represents the number of predictions made by a machine learning algorithm that falls into that particular category

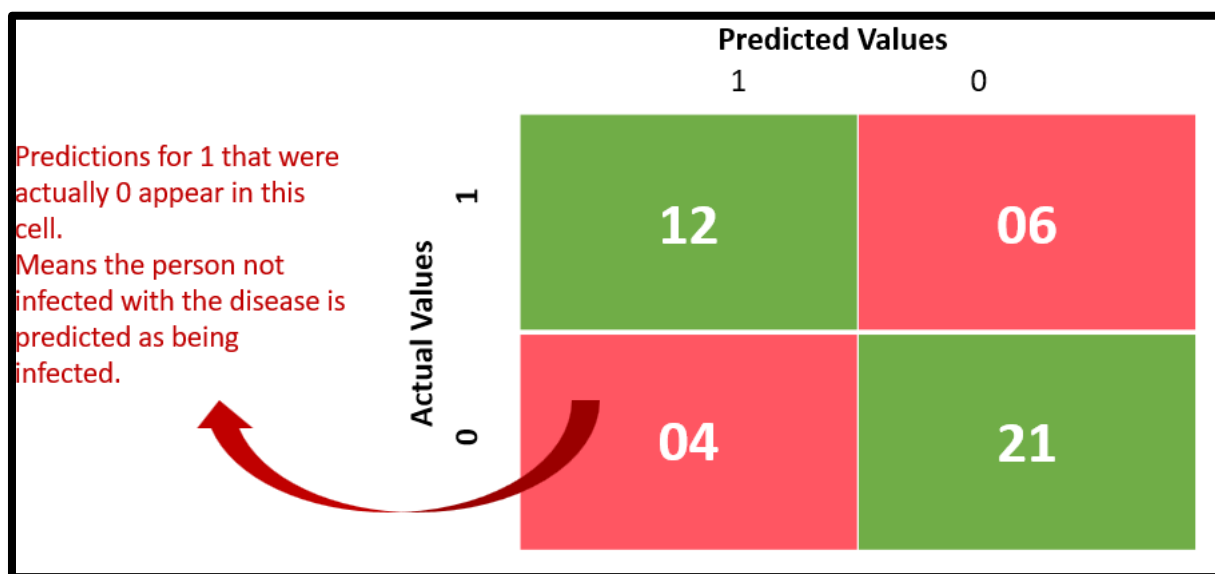
		Predicted Values	
		1	0
Actual Values	1	12	06
	0	04	21

		Predicted Values	
		1	0
Actual Values	1	12	06
	0	04	21

Predictions for 1 that were actually 1 appear in this cell
Means the person infected with the disease is also predicted as being infected.

For example, a machine learning algorithm can predict 0 or 1 and each prediction may actually have been a 0 or 1.





Activity 2: Build the confusion matrix from scratch

Duration: 10 minutes

Purpose: Learn how to create confusion matrix from the scratch.

Say: "The youth need to analyze the situation and tabulate a non-numerical information a numerical one."

Activity Guidelines

- Let's assume we were predicting the presence of a disease; for example, "yes" would mean they have the disease, and "no" would mean they don't have the disease
- So, the AI model will have output is Yes or No
- The following chart shows the actual values and the predicted values
 - Construct a confusion matrix.
 - Can you tell how many are correct predictions among all predictions?

Actual value	Predicted Value
Yes	Yes
No	No
No	Yes
Yes	No
No	No
Yes	Yes
Yes	No
No	No
No	No
No	No

Fill the matrix based on the table given here.

		Predicted Values	
		Yes	No
Actual Values	Yes		
	No		

Count the number of rows having YES in both columns of the table and put the count in the first cell. Similarly, number of rows having YES in the first column and NO in the second column will be shown in the top right cell of confusion matrix. Number of rows having NO in the first column and YES in the second column will be shown in the down left cell of confusion matrix. Lastly, number of rows having NO in the first column and YES in the second column will be shown in the downright cell of confusion matrix.

		Predicted Values	
		Yes	No
Actual Values	Yes	02	02
	No	01	05

Activity Guidelines – Solution

Activity Reflection

- So, there are 07 correct predictions out of 10 predictions.
- What do you think? How good is your model?

Now that you know how to construct a Confusion matrix, let's understand each cell of the matrix in details.

True Positive

- **True Positive (TP)** is the outcome of the model correctly predicting the positive class
- Any class can be assumed as a positive class, and the rest can be assumed as negative
- Let's say class 1 is assumed as the positive class
- Can you tell the TP value from this matrix?

		Predicted Values	
		1	0
Actual Values	1	12	06
	0	04	21

Scenario 1:

Consider you are watching football world cup. Scenario

2:

Consider the earlier example of medical diagnosis of an infected disease.



True Positive examples

- You had predicted that France would win the world cup, and it won.
- In the earlier activity, the cases in which we predicted yes (they have the disease), and they do have the disease.

True Negative

- **True Negative (TN)** is the outcome of the model correctly predicting the negative class.
- Since in the previous example, class 1 is assumed the positive class, class 0 should be assumed the negative class.
- Can you tell the TN value from this matrix?

		Predicted Values	
		1	0
Actual Values	1	12	06
	0	04	21

True Negative examples

- You had predicted that Germany would not win, and it lost
- In the earlier activity, the cases in which we predicted No (they don't have the disease), and they don't have the disease

False Positive

- **False Positive (FP)** is the outcome of the model wrongly predicting the negative class as positive class.
- Here, when a class 0 is predicted as class 1, it falls into the FP cell.
- Can you tell the FP value from this matrix?

		Predicted Values	
		1	0
Actual Values	1	12	06
	0	04	21

False Positive examples

- You had predicted that Germany would win, but it lost.
- In the earlier activity, the cases in which we predicted Yes (they have the disease), and they don't have the disease.

False Negative

- **False Negative (FN)** is the outcome of the model wrongly predicting the positive class as the negative class.
- Here, when class 1 is predicted as class 0, it falls into the FN cell.
- Can you tell the FN value from this matrix?

		Predicted Values	
		1	0
Actual Values	1	12	06
	0	04	21

False Negative examples

- You had predicted that France would not win but it won
- In the earlier activity, the cases in which we predicted No (they don't have the disease), and they have the disease

Accuracy from Confusion matrix

Classification accuracy is the number of correct predictions made as a ratio of all predictions made.

Can you spot the correct predictions?

		Predicted Values	
		1	0
Actual Values	1	12	06
	0	04	21

Calculate the Classification accuracy from this confusion matrix.

Correct predictions=TP+TN

Total predictions=TP+TN+FP+FN

$$\begin{aligned}
 \text{Classification accuracy} &= \frac{\text{Correct predictions}}{\text{Total predictions}} \\
 &= \frac{TP+TN}{TP+TN+FP+FN} \\
 &= \frac{12+21}{12+21+04+06} = 0.767
 \end{aligned}$$

		Predicted Values	
		1	0
Actual Values	1	TP=12	FN=06
	0	FP=04	TN=21

Can we use Accuracy all the time?

- It is only suitable when there are an equal number of observations in each class, i.e., a balanced dataset (which is rarely the case), and that all predictions and prediction errors are equally important, which is often not the case.
- But why is that so? Let's understand it better from the next activity

Activity 3: Calculate the accuracy of the classifier model

Duration: 20 minutes